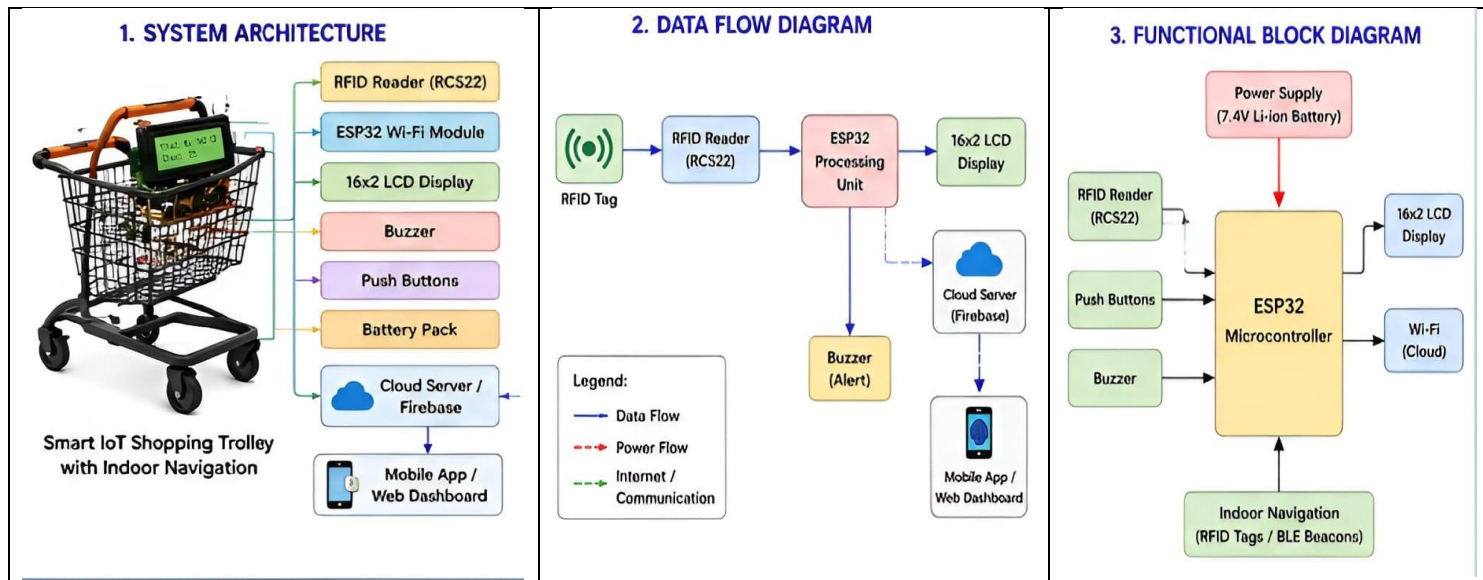


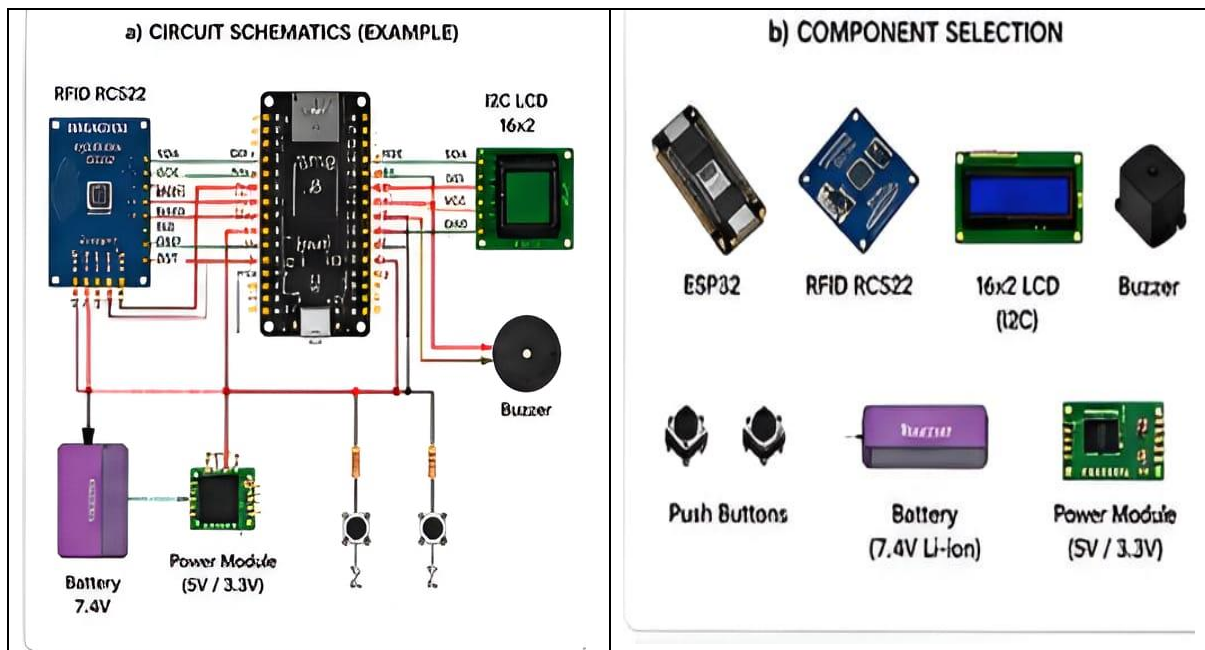
SMART IOT BASED TROLELY USING RFID TAGS

System Architecture



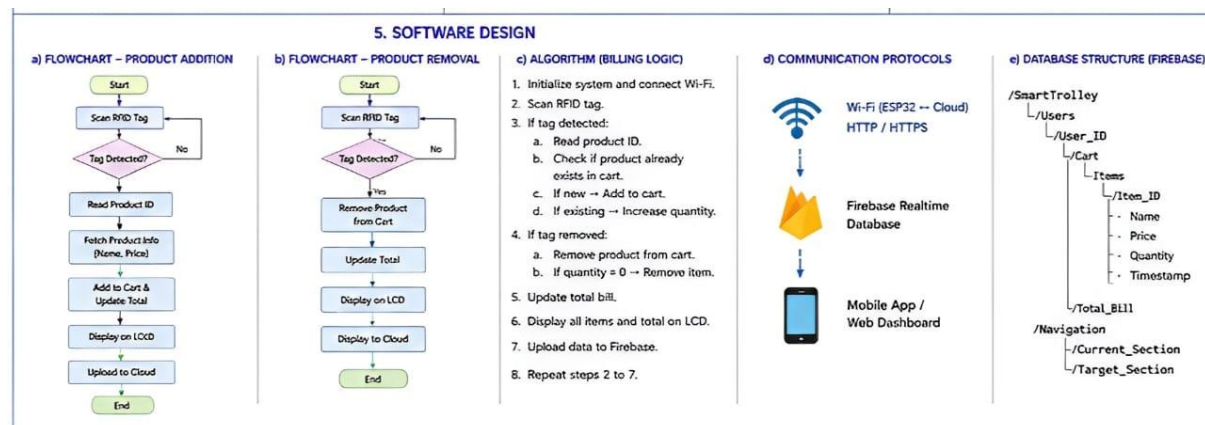
- **Overall Solution Architecture:** Show how the trolley integrates ESP32, BLE, sensors, and cloud storage.
- **Data Flow Diagram:** Illustrate how data moves from sensors → ESP32 → BLE → mobile app/cloud.
- **Functional Block Diagram:** Blocks for power supply, sensors, microcontroller, communication, and storage.
- **Hardware-Software Partitioning:** Define which tasks are handled by hardware (sensing, power regulation) vs software (data processing, BLE communication, database storage).

Hardware Design



- **Final Block Diagram:** ESP32 as the central unit connected to load sensors, RFID scanner, and BLE module.
- **Circuit Schematics:** Show wiring of ESP32 with sensors, display, and power supply.
- **Component Selection:** List chosen sensors (e.g., load cell, IR sensor), ESP32, BLE, battery, display.
- **Bill of Materials (BOM):** Table of components with quantity, cost, and supplier.
- **PCB Layout (if applicable):** Draft layout for compact integration of ESP32 and sensors.

Software Design

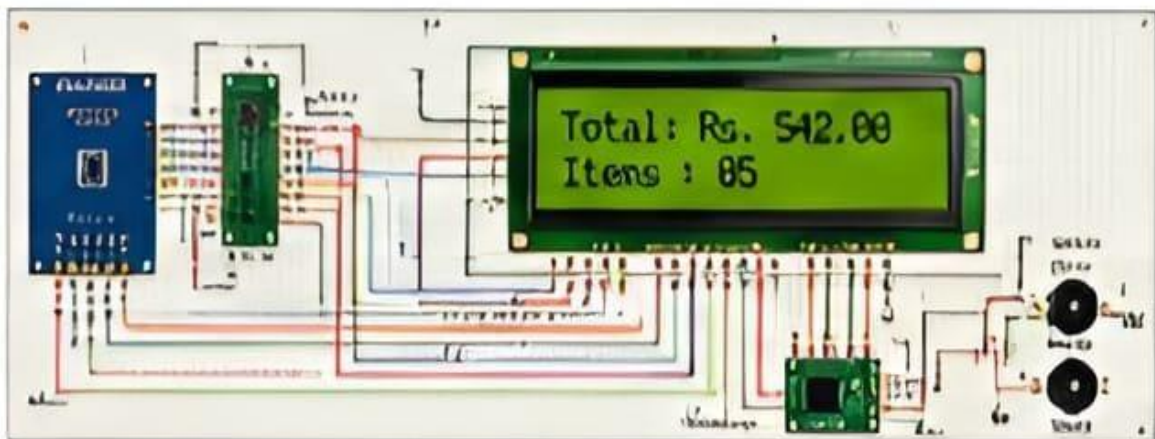


- **Flowcharts:** Logic for scanning items, updating database, and sending data via BLE.
- **Algorithms:** Item recognition, weight calculation, and billing automation.
- **Communication Protocols:** BLE protocol for mobile app connectivity, possibly MQTT/HTTP for cloud sync.
- **Database Structure (if applicable):** Tables for item ID, weight, price, and transaction logs.

Simulation

6. SIMULATION

a) PROTEUS SIMULATION



- **Proteus Simulation:**
 - Use **Proteus** for ESP32 + sensor simulation.
- **Initial Results:** Show sensor readings, BLE connectivity test, and sample data storage output.